

COMP 541 Final Report: Produce Department Analysis

Instacart Training-Order Dataset Behavior Study

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This report summarizes the final analysis of the Produce department in the Instacart training-order dataset. The goal was to combine basic dataset information, statistical analysis, frequent pattern mining, and an extra-credit Produce shopping style clustering analysis into one clear narrative.

1. Dataset and Preparation

The project focused on the Produce department because it is broad enough to show meaningful customer behavior while still being manageable for analysis. Using departments.csv, Produce was identified as department_id = 4. The products table was filtered to keep Produce only, then joined with aisles, training orders, and order-level metadata. This produced a clean Produce transaction table with 1,684 Produce products, 409,087 order-product rows, and 96,927 Produce baskets/orders. The merge quality check showed 0 missing user_id values, so the prepared dataset was suitable for both statistical analysis and FP-Growth mining.

Prepared dataset summary	Value
Produce products	1,684
Produce order-product rows	409,087
Produce baskets/orders	96,927
Missing user_id after merge	0

2. Statistical Findings

The first three analyses described the structure of Produce, the most purchased products, and the timing of Produce demand.

2.1 Produce Products by Aisle

Produce products were concentrated in a few main aisles. The largest aisle was packaged vegetables fruits with 615 products (36.52%), followed by fresh vegetables with 569 products (33.79%), and fresh fruits with 382 products (22.68%). Together, the top two aisles account for over 70% of all Produce products. This shows that the department is centered around a small number of core fruit and vegetable categories.

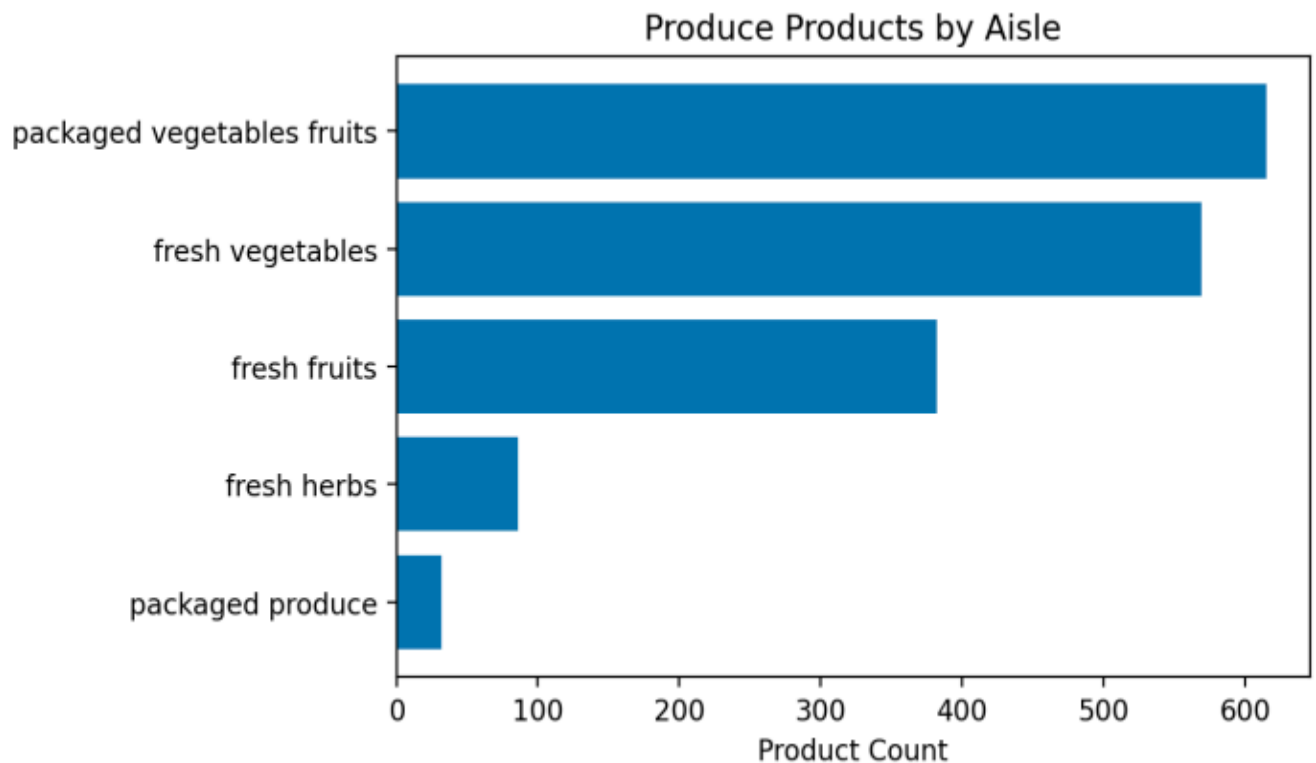


Figure 1. Distribution of Produce products across Produce aisles.

2.2 Top 10 Produce Products

Produce demand was dominated by a small set of staple products. Banana ranked first with 18,726 purchases (19.55% of the top-10 total), followed by Bag of Organic Bananas with 15,480 purchases (16.16%). Other top products included Organic Strawberries, Organic Baby Spinach, Large Lemon, and avocado products. These results suggest that Produce demand is highly concentrated around a small group of everyday items.

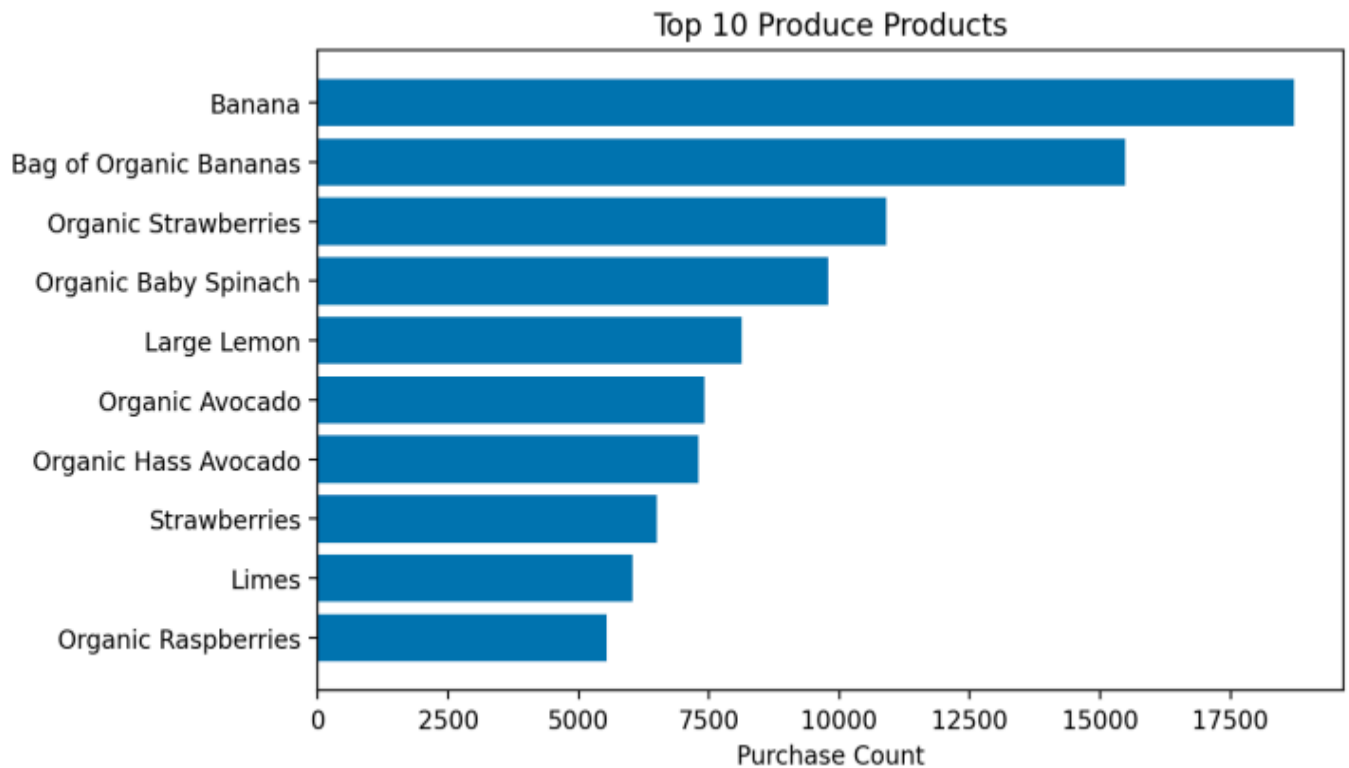


Figure 2. Ten most frequently purchased Produce products in the training orders.

2.3 Produce Purchase Timing

Produce purchases also showed clear timing patterns. By day of week, the highest purchase count occurred on `order_dow = 0`, with `order_dow = 1` and `order_dow = 6` also showing strong activity. By hour of day, Produce purchases were very low during late night and early morning hours, increased quickly after 7:00, stayed high through late morning and the afternoon, and peaked around 14:00-15:00. This indicates that Produce shopping is closely tied to normal daytime grocery trips rather than overnight ordering behavior.

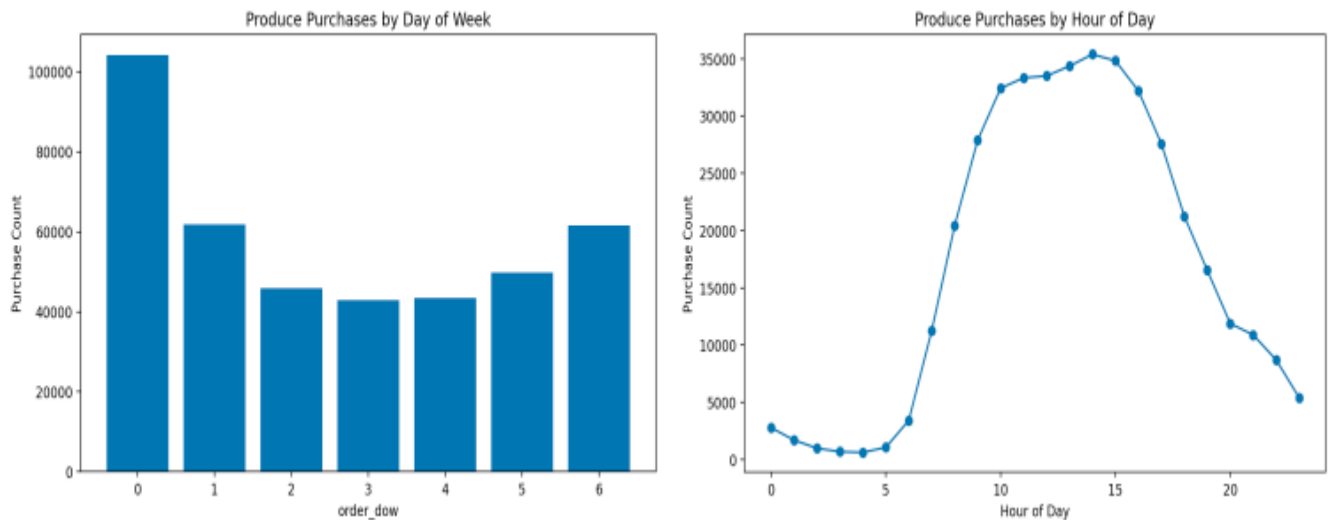


Figure 3. Produce purchases by day of week and by hour of day.

3. Frequent Pattern Mining Results

FP-Growth was used for all pattern-mining analyses because it is efficient for large transaction data and fits the project requirement better than manual pair counting. Three rule metrics were used throughout the mining steps: support count, support, and confidence. Thresholds were adjusted by analysis level to keep the rules meaningful without allowing too many rare or weak patterns.

3.1 Product-to-Product Rules

For product-level mining, Produce baskets were built from individual orders. This created 96,927 baskets, yielded 345 frequent itemsets, and after filtering left 217 one-to-one rules. The thresholds were support count ≥ 500 and confidence ≥ 0.10 . The strongest rules showed that bananas served as a central product in Produce co-purchase behavior. Examples include Organic Strawberries $\rightarrow$ Bag of Organic Bananas (support count 3,074, support 0.0317, confidence 0.2822) and Organic Hass Avocado $\rightarrow$ Bag of Organic Bananas (support count 2,420, support 0.0250, confidence 0.3318). Overall, product-level FP-Growth showed that bananas, berries, avocados, spinach, and lemons formed recurring Produce purchase combinations.

Top product-to-product rules in Produce	Support Count	Support	Confidence
Organic Strawberries $\rightarrow$ Bag of Organic Bananas	3074	0.0317	0.2822
Organic Hass Avocado $\rightarrow$ Bag of Organic Bananas	2420	0.0250	0.3318
Organic Baby Spinach $\rightarrow$ Bag of Organic Bananas	2236	0.0231	0.2285
Organic Avocado $\rightarrow$ Banana	2216	0.0229	0.2991
Large Lemon $\rightarrow$ Banana	2158	0.0223	0.2653

3.2 Aisle-to-Aisle Rules

At the aisle level, the same 96,927 Produce baskets were converted to aisle baskets. This mining step produced 23 frequent aisle itemsets and 15 filtered aisle rules. Because aisle patterns are broader and more frequent than product rules, the support threshold was made stricter at support count ≥ 1000 , while confidence remained 0.10. The strongest aisle rule was fresh vegetables $\rightarrow$ fresh fruits (support count 42,949, support 0.4431, confidence 0.7258). Other strong rules showed that packaged vegetables fruits are tightly connected with both fresh fruits and fresh vegetables. This reveals broader department structure than the product-level rules and shows that Produce shopping is often built from multiple major aisle categories.

Top aisle-to-aisle rules in Produce	Support Count	Support	Confidence
fresh vegetables $\rightarrow$ fresh fruits	42949	0.4431	0.7258
fresh fruits $\rightarrow$ fresh vegetables	42949	0.4431	0.5950
packaged vegetables fruits $\rightarrow$ fresh fruits	37694	0.3889	0.7500
fresh fruits $\rightarrow$ packaged vegetables fruits	37694	0.3889	0.5222
packaged vegetables fruits $\rightarrow$ fresh vegetables	33112	0.3416	0.6589

3.3 Cross-Department Rules

For the cross-department analysis, full training orders were converted into department baskets. This created 131,209 training orders, of which 96,927 contained Produce. The mining step yielded 169 frequent department itemsets and 16 filtered Produce → department rules. The thresholds were support count ≥ 1000 and confidence ≥ 0.05 . Confidence was set lower here because cross-department rules are broader and naturally weaker than Produce-only rules. The strongest rule was produce → dairy eggs (support count 71,356, support 0.5438, confidence 0.7362), followed by beverages, snacks, frozen, and pantry. These findings show that Produce is not purchased in isolation; it is strongly connected to the broader grocery basket.

Top cross-department rules involving Produce	Support Count	Support	Confidence
produce → dairy eggs	71356	0.5438	0.7362
produce → beverages	44893	0.3421	0.4632
produce → snacks	44389	0.3383	0.4580
produce → frozen	41264	0.3145	0.4257
produce → pantry	39075	0.2978	0.4031

4. Extra-Credit Analysis: Produce Shopping Style Clusters

The extra-credit step used K-means clustering to identify different shopping styles among orders that contained Produce. This goes beyond the required statistical summaries and pattern-mining analyses because it groups orders based on their overall basket behavior rather than only counting products or mining co-purchase rules. For each Produce-containing order, I used features such as the number of Produce items in the basket, the total number of items in the order, the number of unique departments, the number of unique Produce aisles, and the share of Produce within the full basket.

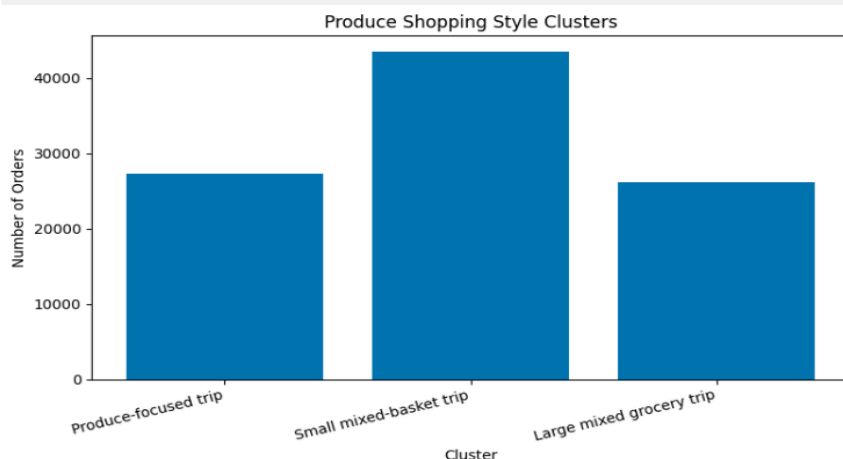
The clustering results revealed three main Produce shopping styles. The first cluster represents Produce-focused trips, where Produce makes up the largest share of the basket. The second cluster represents small mixed-basket trips, where customers buy a small amount of Produce as part of a broader order. The third cluster represents large mixed grocery trips, where customers buy many items across many departments and also include a substantial amount of Produce. The largest group was the small mixed-basket trip cluster, accounting for 44.86% of Produce-containing orders, followed by Produce-focused trips at 28.16% and large mixed grocery trips at 26.97%. These findings are meaningful because they show that Produce does not play only one role in customer shopping behavior. Instead, it appears in targeted Produce-heavy trips, smaller mixed trips, and larger full-basket grocery trips.

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... Preview of Produce order features:
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	order_id	produce_item_count	total_items_in_order	unique_departments	unique_produce_aisles	produce_share
0	1	4	8	3	2	0.5000
1	36	3	8	4	2	0.3750
2	38	6	9	4	3	0.6667
3	96	4	7	3	2	0.5714
4	98	7	49	12	4	0.1429

Produce shopping style cluster summary:

	cluster	cluster_label	produce_item_count	total_items_in_order	unique_departments	unique_produce_aisles	produce_share	order_count	percentage_of_orders
0	0	Produce-focused trip	4.06	6.79	3.02	2.22	0.63	27299	28.16
1	1	Small mixed-basket trip	2.06	10.21	5.69	1.44	0.22	43483	44.86
2	2	Large mixed grocery trip	7.99	21.87	7.80	3.10	0.38	26145	26.97



5. Conclusion

Overall, the Produce department emerged as one of the most connected and informative areas in the dataset. It is concentrated in a few dominant aisles, driven by a small group of staple products, strongly tied to daytime shopping behavior, and deeply linked to the broader grocery basket through both association rules and broader shopping-style behavior. The statistical analyses showed that Produce demand is concentrated in a few key aisles and products, while the pattern-mining analyses revealed meaningful product-level, aisle-level, and cross-department relationships. In addition, the extra-credit clustering analysis showed that Produce appears in multiple shopping styles, including Produce-focused trips, small mixed-basket trips, and large mixed grocery trips. These findings suggest several practical implications: maintain strong inventory coverage for staple Produce items, use Produce-centered bundles for cross-selling, align promotions with peak shopping hours, improve recommendations across related departments, and support different shopping styles in basket-level marketing strategies. In short, Produce is not only a department customers buy from. It is a core category that helps shape routine grocery behavior across the larger household basket.